Harry Zhang

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Education

University of Toronto, BASc in Computer Engineering, PEY Co-op — Sept 2024 – Apr 2029

- **cGPA:** 3.65/4.00
- **Coursework:** Programming Fundamentals, Computer Organizations, Electronics, Digital Systems, Signals and Systems, Linear Algebra, Calculus I, II & III

Experience

A Living Lab for Primary Care Physicians at Work (Undergraduate Researcher) — Jan 2025 – Apr 2025

Wellness and Health Enhancement Engineering Lab (WHEEL), University of Toronto

- Supervised by Dr. Enid Montague, collaborated with UofT's Wellness and Health Enhancement Engineering (WHEEL) Lab to redesign a usability testing lab for studying physician-technology interactions
- Tested and validated Livestreamer's prototype data capture system by executing multimodal performance tests, achieved strong multimodal coverage (0.91).
- Coordinated quality assurance of 3D lab layout models using Fusion 360 and Blender, improved communication with clients and enabled accurate assessment of feasibility.

Projects

Geographic Information System (GIS) for City of Toronto — Jan 2026 – Apr 2026

ECE297 Software Design Project, University of Toronto

- Applied **Git/GitHub** and **VS Code debugger** to manage iterative development, debug logic/memory issues, and validate performance, ensured stable map loading and efficient query execution
- Developed modular C++ solutions using **STL containers and algorithms** to process OSM data; wrote clean, maintainable code and reduced runtime errors during large-scale dataset testing.
- Wrote clear technical documentation and researched OpenStreetMap data structures by studying the API and analyzing map data formats, reduced implementation errors and kept the team aligned during development.

Connect8 Puzzle Game on FPGA — Nov 2025

ECE241 (Digital Systems) Course Project, University of Toronto

- Built a full hardware puzzle game in Verilog on the DE1-SoC by implementing block movement, collision detection, and an 8x8 line-clear engine, produced a stable core game loop that ran entirely on hardware.
- Integrated a PS/2 keyboard controller, **VGA renderer**, and a 16-bit **LFSR block generator** to support real-time input and display, gave the game smooth controls and a consistent 60 FPS output.
- Debugged critical rotation and collision handling errors and revised code based on testbench validation

Skills

Programming: C, C++, Python, RISC-V Assembly (NIOS V), MATLAB, Verilog

Technologies: VS Code, Git/Github, AutoCAD

Honors & Awards

Dean's List	Fall 2024, Winter & Fall 2025
Faculty Of Applied Science And Engineering Admission Scholarship	$10,000
The Edward S. Rogers Sr. Department Admission Scholarship	$2,500